Satisfaction, Stress & the Evidence Trail

A school-enterprise employee well-being roundtable

How satisfaction and dissatisfaction become trackable research evidence

THE ROUNDTABLE QUESTION

How can employee satisfaction or dissatisfaction become trackable research evidence?

No organization-level conclusions

We track evidence, not verdicts about the organization.

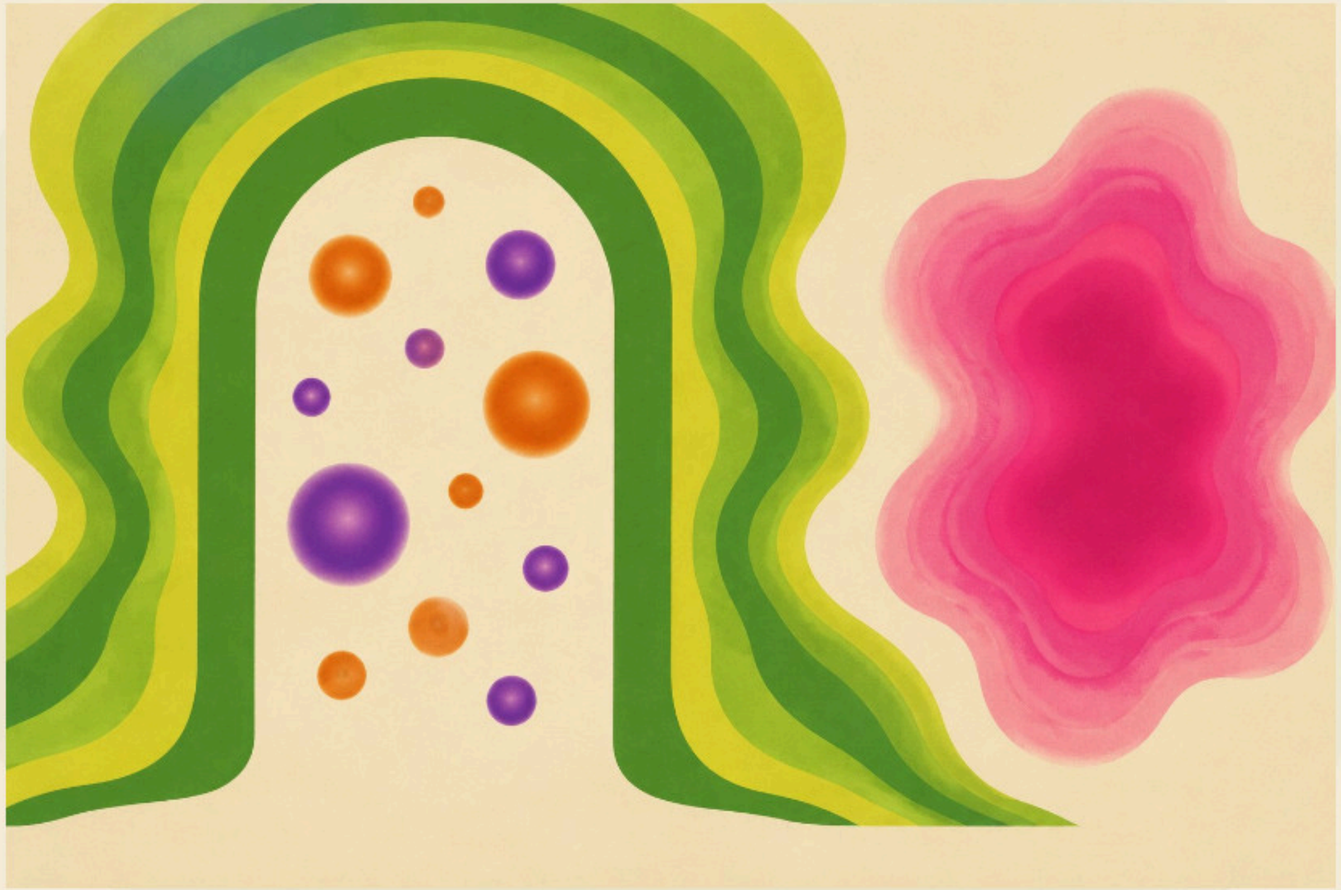
No clinical diagnosis

This is research, not clinical assessment.

Evidence over impressions

Satisfaction and dissatisfaction become data, not anecdotes.

What we study — and what we leave outside



INSIDE — WHAT WE STUDY

OUTSIDE — CLINICAL DIAGNOSIS

No clinical diagnosis will be made in this study.

- Work demands
- Satisfaction expressions
- Psychological stress cues (self-reported, non-clinical)

- Depression
- Anxiety disorders
- Other clinical conditions (not studied here)

Variable — a thing we measure or observe that can change from person to person or over time.

What the research agrees on

PUBLIC LITERATURE · CONSENSUS

Job satisfaction is linked to well-being

A **modest but consistent** positive association — people who report higher job satisfaction tend to report better well-being and mental health.

How strong is the link?

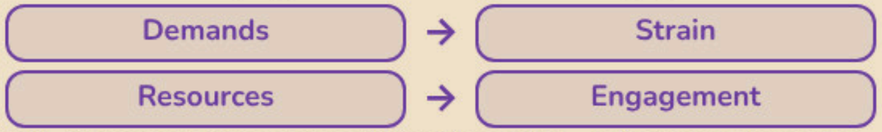
r ≈ 0.23 overall; **r ≈ 0.26** for mental health; **r ≈ 0.19** for physical health.

Faragher et al. (2005), meta-analysis of 126 studies.

Meta-analysis = a study that statistically combines results from many separate studies. **Effect size (r)** = how strong the association is, from 0 (none) to 1 (perfect); 0.2–0.3 is a small-to-moderate link.

What explains it? The JD-R model

Work **demands** (workload, pressure) drive strain and burnout; work **resources** (support, autonomy) drive engagement.



Job Demands-Resources (JD-R) model = a framework saying work conditions shape how people feel: too many demands wear people down, good resources keep them engaged (Demerouti et al., 2001; Bakker & Demerouti, 2017).

What "consensus" means here

The association is **robust** — it shows up across many studies. But consensus on **association** is not consensus on **causation**: the literature does not yet settle which direction the link runs.

Where the research still argues

The consensus is real — but four debates stay **open**. Each one is a reason to be careful about what we claim.

DEBATE 01

Which way does the arrow point?

Does job satisfaction improve health — or does better health make people report higher satisfaction? Studies suggest the link runs *both ways*; the direction is still contested.

Causality — which variable actually changes the other, not just whether they move together.

DEBATE 02

Are the ratings trustworthy?

People who rate one thing highly tend to rate others highly too. When satisfaction and well-being come from the *same questionnaire*, the link can look stronger than it really is.

Self-report bias — answers shaped by how people present themselves. **Common method variance** — extra similarity between measures simply because they share one source.

DEBATE 03

Do scales mean the same thing everywhere?

A score of 4 out of 5 may mean something different across cultures or groups. Comparisons across groups are only fair if the scale behaves the same way in each.

Measurement invariance — the property that a scale measures the same thing, in the same way, across different groups.

DEBATE 04

What exactly is “satisfaction”?

Is job satisfaction the same thing as happiness or well-being? Researchers disagree about whether these are one construct or several overlapping ones.

Construct validity — whether a measure actually captures the concept it claims to measure.

- These open debates decide which links in our variable map can be **tested** — and which stay **interpretive hypotheses only** (pages 18–22).

So what does that mean for us?

STREAM A · PUBLIC-LITERATURE SCALES

Consensus gives us yardsticks

Validated scales we can reuse — shared measures of satisfaction and stress, already tested by researchers.

STREAM B · PLANNED OBSERVATION & INTERVIEWS

Debates tell us where to be careful

Our own data must be collected and interpreted with the literature's limits in mind.



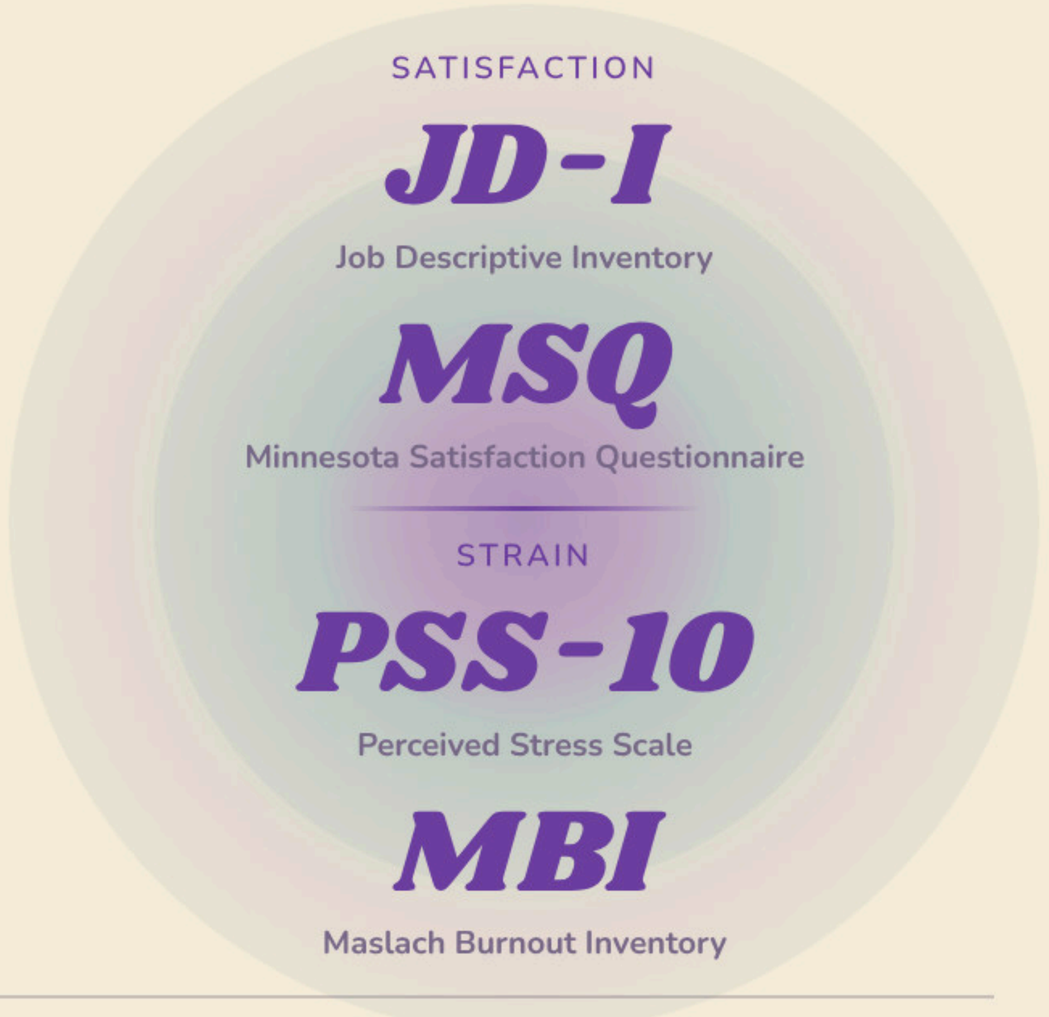
Two streams → one variable map

Every measure joins the same trail of variables.

STREAM A · RESEARCH EVIDENCE

Stream A: the literature's yardsticks

Public research hands us ready-made yardsticks: **scales** — standardized sets of questions with a scoring rule, used the same way everywhere. Because they are **validated**, the numbers we collect can be compared directly with published studies.



Scale — a fixed set of questions plus a scoring rule, administered identically everywhere.

Indicator — a measurable sign of something we cannot see directly, like satisfaction or strain.

Validated — tested and shown to measure what it claims to measure.

STREAM A · RESEARCH EVIDENCE

Measuring satisfaction

Facet — one sub-part of what a scale measures.
Likert scale — a rating scale, e.g. 1 = strongly disagree to 7 = strongly agree.

JD-I

Job Descriptive Inventory (Spector, 1985)

MEASURES	Overall job satisfaction, split into 5 facets: work itself, supervision, pay, promotion, coworkers.
ITEMS	44 items — 9 work, 9 supervision, 9 pay, 8 promotion, 9 coworkers.
SCORING	7-point scale (1 = strongly disagree … 7 = strongly agree). About half the items are reverse-worded and must be reverse-scored; facet scores are averages of their items.

IN PLAIN LANGUAGE

You rate how you feel about five concrete parts of your job — the work, your boss, your pay, your chances of promotion, and your coworkers.

MSQ

Minnesota Satisfaction Questionnaire (Weiss, Dawis, England & Lofquist, 1967)

MEASURES	Job satisfaction across 5 facets: achievement, variety, independence, recognition, security.
ITEMS	Long form: 100 items (20 per facet). Short form: 20 items (4 per facet).
SCORING	5-point scale (1 = very dissatisfied … 5 = very satisfied). Facet scores are sums or averages of the items in each facet.

IN PLAIN LANGUAGE

You rate how satisfied you are with five things your job can give you — achievement, variety, independence, recognition, and job security.

STREAM A · RESEARCH EVIDENCE

Measuring strain

Subscale — one sub-part of what an inventory measures. **Time window** — the period the questions refer to.

PSS-10

Perceived Stress Scale (Cohen, Kamarck & Mermelstein, 1983)

MEASURES	Perceived stress — how unmanageable, unpredictable, uncontrollable, or piling-up life situations feel.
ITEMS	10 items (14- and 4-item versions also exist).
TIME WINDOW	The past month.
SCORING	Each item rated 0–4 (0 = never … 4 = very often). 4 items are worded positively and must be reverse-scored. Total 0–40; higher = more perceived stress.

IN PLAIN LANGUAGE

You answer ten questions about how often things in your life felt stressful over the past month — it measures how overwhelmed you feel, not whether you have a mental illness.

MBI

Maslach Burnout Inventory (Maslach & Jackson, 1981)

MEASURES	Burnout — a work-related syndrome of emotional exhaustion, depersonalization, and reduced personal accomplishment.
ITEMS	22 items — 9 emotional exhaustion, 5 depersonalization, 8 personal accomplishment.
TIME WINDOW	Current work experience (frequency of feelings).
SCORING	7-point frequency scale (0 = never … 6 = every day). Subscale scores are sums of their items.

IN PLAIN LANGUAGE

You say how often you feel emotionally drained, how detached or cynical you feel about your work, and how effective you feel at it.

BOUNDARY CHECK

Both instruments measure **perceived strain** — self-reports of stress and exhaustion. They are **not clinical diagnoses**; the MBI is not a diagnostic instrument.

STREAM B · EMPLOYEE VOICE

Stream B: what we will observe and ask

> Public scales tell us what **others** found. Our own data will tell us what is happening **here**.

Structured observation

Watching work situations with a fixed set of rules, so every observation is recorded the same way.

Semi-structured interview

A conversation around a fixed question set — the same questions, but room for people to say more.

Coding

Sorting interview answers into agreed categories so they can be counted and compared.

Work demands

How heavy and how interrupted the work is — observed on site.

Satisfaction expressions

What people say about their work — heard in interviews.

Psychological stress cues

Self-reported strain — fatigue, sleep, concentration.

STREAM B · EMPLOYEE VOICE

Work demands: what we observe

A fixed set of work situations, observed the same way each time — so demands become comparable across people and over time.

Workload

Task volume and pace — how much work arrives, and how fast it must be done.

Interruptions

Frequency of task switching — how often work is broken off to start something else.

Autonomy

Choice over how work is done — how much control the person has over their own tasks.

The observation protocol

Structured observation — a fixed set of situations observed the same way each time.

WHAT	WHEN	HOW LONG
Workload, interruptions, autonomy	During routine work situations	Same duration every session
Same situations, same rules	Repeated over time	Repeated for each person

A trained observer records with the same rules each time — nothing is invented, nothing is skipped.

STREAM B · EMPLOYEE VOICE

Satisfaction expressions: what we hear

Interview language becomes data: what people say, how often, and in what tone — coded with the same rules for every interview.

Words

The actual expressions of satisfaction or dissatisfaction people use in interviews.

Tone

How the words are delivered — positive, negative, or mixed in feeling.

Frequency

How often positive or negative expressions come up — counted, not guessed.

The coding scheme

Semi-structured interviews — a fixed question set, room to say more; transcripts coded thematically.

WHAT COUNTS	HOW IT IS COUNTED
Any expression of satisfaction or dissatisfaction	Each expression is marked once in the transcript
Positive vs negative tone	Marked per expression, same rules every time
Recurring themes	Grouped thematically across interviews

Thematic coding — sorting answers into agreed themes, so every interview is read with the same eyes.

STREAM B · EMPLOYEE VOICE

Stress cues: what people report

Self-reported strain — the same questions each time, in interviews and on short checklists.

Self-report

People telling us about their own experience — their words, not a measurement taken from them.

Cue

A signal worth tracking — not a verdict, and not a diagnosis.

Checklist

A short list of questions people answer about themselves — the same list each time.

BOUNDARY CHECK

These are **cues, not symptoms of a disorder** — they stay inside the boundary set on page 3.

The self-report checklist

Short questions, answered by the person themselves — the same questions each time.

☐ **Fatigue**	Feeling worn out by work
☐ **Sleep trouble**	Difficulty falling or staying asleep
☐ **Difficulty concentrating**	Struggling to focus on tasks

Interview questions plus this checklist — repeated so strain can be tracked over time.

Two streams, one evidence trail



STREAM A literature scales

STREAM B interview observations

ONE EVIDENCE TRAIL

The two streams measure different things but feed the same variable map.

Merging the streams a variable map

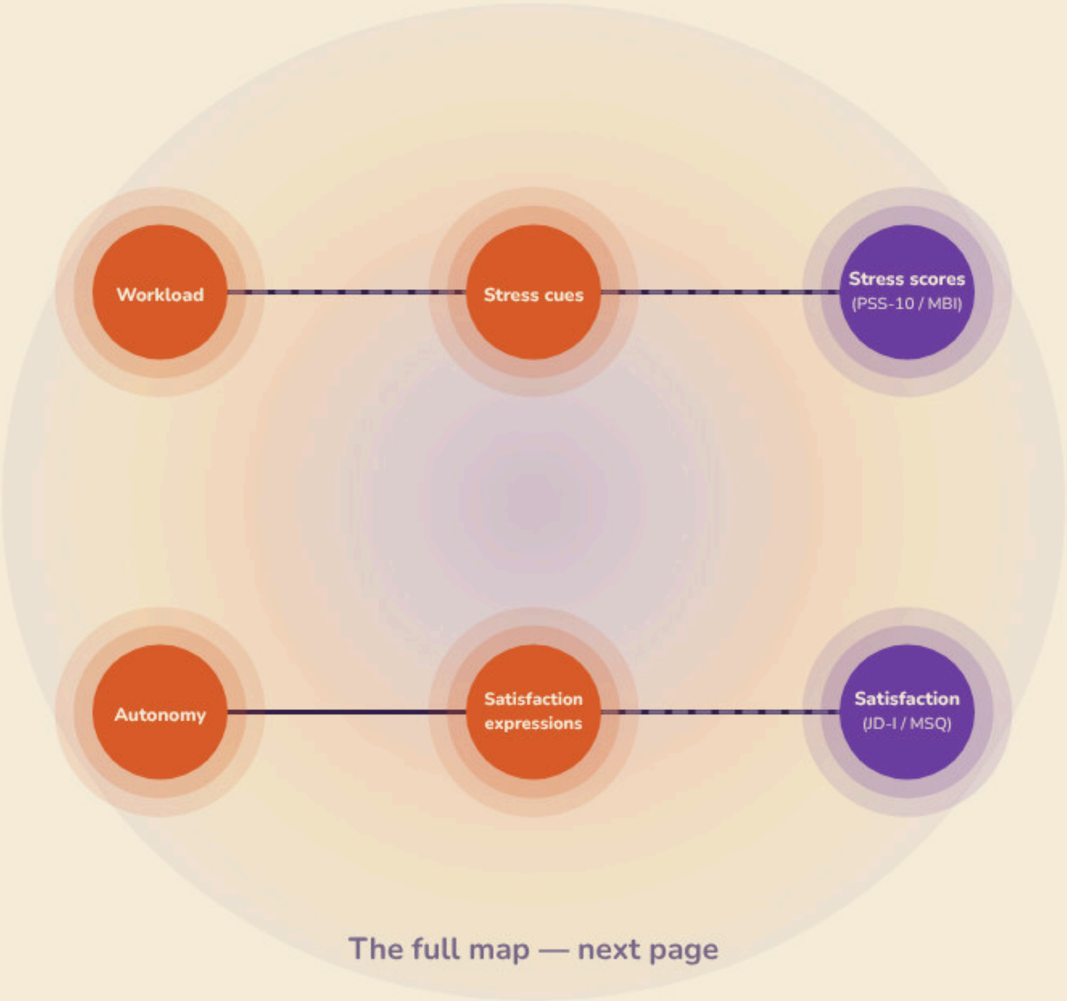
A list shows **what** we measure. A map shows **how the measures relate** — so satisfaction and stress become trackable evidence, not just a checklist.

Variable
Anything we measure or observe — workload, a stress score, the tone of an interview.

Node
A variable drawn as a circle on the map.

Edge
A line connecting two nodes on the map.

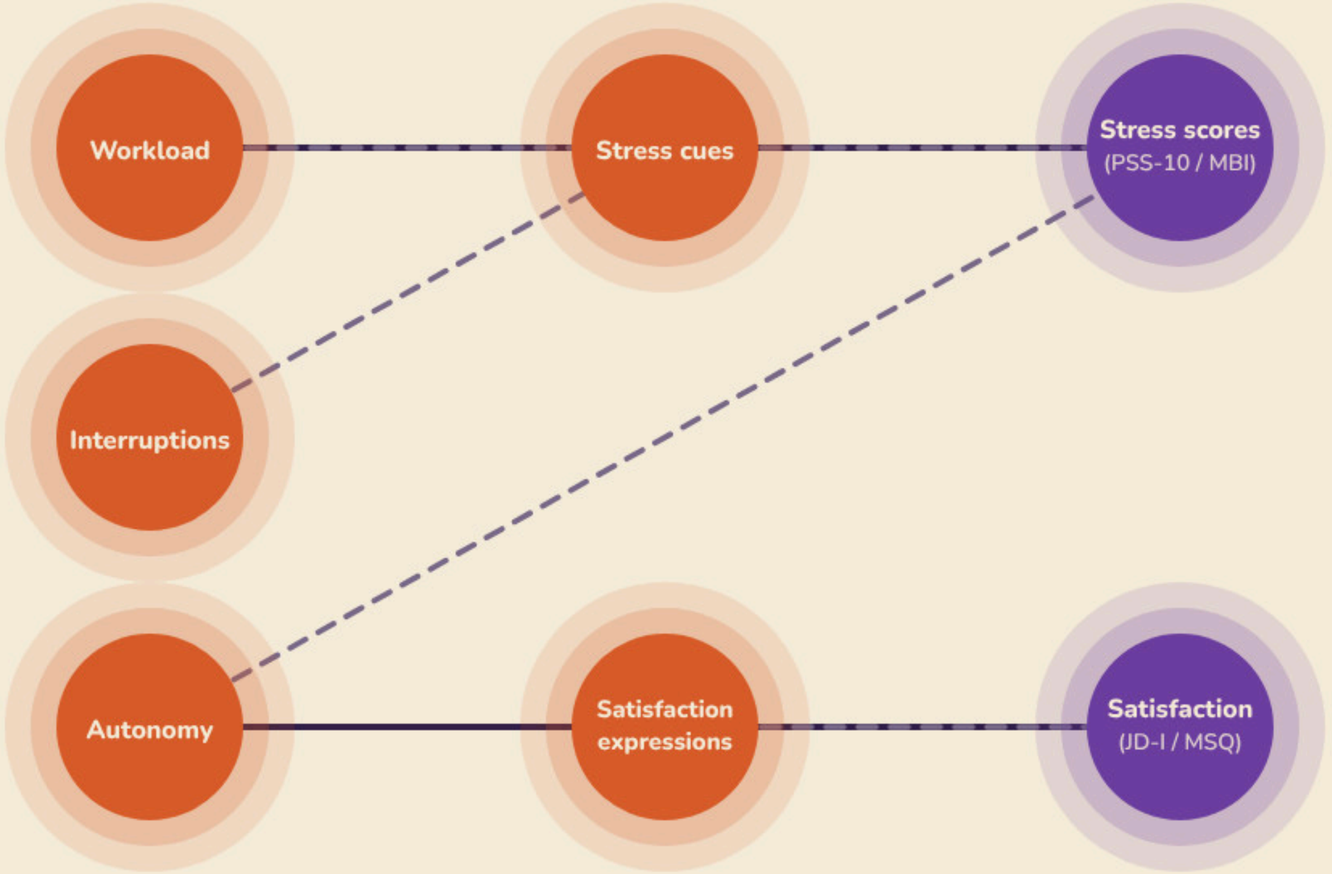
Link
The relationship an edge stands for: a proposed connection between two variables.



The full map — next page

The variable map

Both streams on one map — violet from the literature's yardsticks, orange from our own observations.



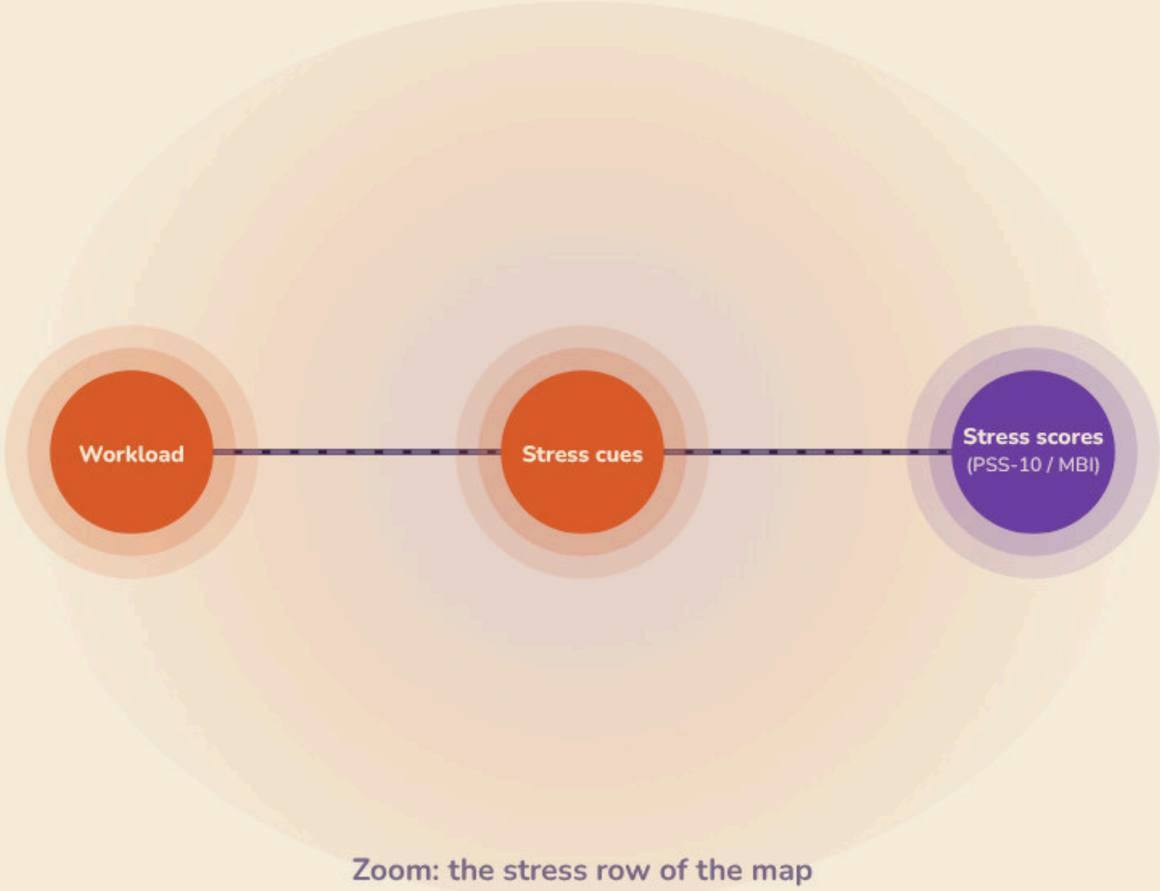
How to read the map

Violet node
Stream A — a variable from the literature's yardsticks (published scales).

Orange node
Stream B — a variable from our own observations and interviews.

Testable link
A relationship we can check with correlation or regression.

Interpretive hypothesis
A proposed relationship we can only reason about — not statistically test.



Zoom: the stress row of the map

A link is a proposed relationship — not a proven cause.

Two kinds of links

Some links on the variable map we can measure. Others we can only interpret.

Testable QUANTIFIABLE

Testable — a link whose strength we can put a number on with **correlation** (how tightly two scores move together) or **regression** (how much one score predicts another).



SATISFACTION SCORE ↔ STRESS SCORE — Two scale scores — correlation gives a number.

WORKLOAD → STRESS SCORE — Observed demand vs. strain indicator — regression can quantify it.

Interpretive HYPOTHESIS ONLY

Interpretive — a link we can only read as a hypothesis: the connection depends on context, **confounding** (other factors moving along with both variables), or limits of our measures.



AUTONOMY → SATISFACTION EXPRESSION — Does more autonomy change how people express satisfaction? Context-dependent.

INTERRUPTIONS → STRESS CUES — Interview cues are soft measures — a hypothesis, not a number.

Why some links stay interpretive

Confounding

Other factors shift with both variables, so the link may be indirect.

Measurement limits

Interview cues and self-reports are softer than scale scores.

Small samples

Few observations make any number too unstable to trust.

Context dependence

The link may hold in one team or role and not another.

Correlation

How two variables move together.

When one goes up, does the other go up too — or down?

> **Not causation.**
> Two things moving together does not mean one causes the other.

−1 ——— +1

opposite directions 0 = no relationship move together

Correlation — a single number describing how tightly two variables move together.

Scatter plot — one dot per person; the cloud's tilt shows the relationship.

Coefficient — that number, from −1 to +1; the steeper the tilt, the closer to ±1.



Illustrative, not data — each dot is one person; the cloud's tilt is the correlation.

Regression

One step further than correlation. It predicts one variable from others — and can hold some variables steady.

> **What it adds: prediction + control.**
> Correlation says "they move together". Regression says "if we know one, we can estimate the other — even while holding others steady."

"Holding **workload** constant, does satisfaction still relate to stress?"

Regression — a model that fits a line (or curve) through the cloud to make predictions.

Prediction — using the line to estimate a value you haven't measured yet.

Holding constant — "freezing" another variable so it can't explain the relationship you're testing.



Illustrative, not data — the line shows the predicted relationship through the cloud.

Some links stay interpretive

Real hypotheses — but we describe them, we do not quantify them.



Wavy, not taut — unlike the solid links of correlation and regression.

Why they stay interpretive

- ≈ **Confounding** — a third factor could explain both sides.
- ≈ **Measurement limits** — self-report only.
- ≈ **Small samples** — too few people to test reliably.
- ≈ **Context dependence** — the link may hold here but not elsewhere.

> **In practice: we describe the relationship, we do not quantify it.**

Interpretive hypothesis — a plausible link we can talk about but not measure with current data.

Confounding — a hidden third variable that could be the real driver of both.

Context dependence — a relationship that may change with the setting, team, or time.

Taut links, wavy hypotheses

The map's **solid** and **dashed** edges become currents — taut where testable, wavy where interpretive.

Interpretive hypotheses stay wavy

Measurable links tighten

Every term, in plain language

Keep this page. Every technical word in this deck, in words a frontline manager can use today.

Variable — anything we measure or track: workload, satisfaction, sleep.

Node — a dot on the variable map; each dot is one variable.

Edge — a line between two dots, showing a possible connection.

Link — a relationship between two variables we want to test.

Scale — a standard questionnaire that measures one thing the same way for everyone.

Indicator — a single question or sign that points to a bigger thing, like strain.

Validated — proven in earlier studies to actually measure what it claims to.

Facet — one side of a bigger concept, e.g. workload is a facet of job demands.

Likert scale — a rating from "strongly disagree" to "strongly agree".

Correlation — two things move together; it does not prove one causes the other.

Coefficient — a number from −1 to +1 showing how tightly two things move together.

Scatter plot — a chart of dots, one per person, showing how two measures line up.

Regression — a test of how much one thing changes when another changes.

Prediction — estimating a likely outcome from what we already measure.

Holding constant — ignoring other factors so we can see one link clearly.

Interpretive hypothesis — a sensible guess we can discuss but not statistically test.

Confounding — a third factor that could explain the link we see.

Context dependence — a finding that may only hold in one setting, not everywhere.

Structured observation — watching work with a fixed list of things to note down.

Semi-structured interview — a conversation with set questions, but room to follow up.

Thematic coding — sorting interview answers into recurring themes.

Frequency — how often something happens, e.g. interruptions per hour.

Self-report — what people tell us about their own experience.

Cue — an observable sign, like a sigh or a pause, that may hint at strain.

Checklist — a short list of signs to tick off while observing.

Threshold — a set level that, when crossed, triggers a review.

Anonymization — removing names so no one can be identified.

Aggregation — combining answers into group totals, never individual scores.

Meta-analysis — pooling results from many studies to see the overall picture.

Effect size — how big a difference or link really is, not just whether it exists.

Job Demands–Resources model — the idea that heavy demands plus few resources lead to strain.

Measurement invariance — checking that a scale means the same thing across groups.

Construct validity — does the tool actually measure the concept it claims to?

Common method variance — fake links that appear because everything was measured the same way.

Ethics & consent

The conditions under which the evidence streams on pages 7–13 can operate.

"We ask first. We never push."

01 Informed consent

Participants know what we collect, why, and how it is used — before they agree.

02 Voluntary participation

No one is required to take part. Saying no changes nothing.

03 Right to withdraw

Participants can stop at any time, without consequence.

04 Confidentiality

Individual responses are not shared outside the research team.

What we will not do

The negative side of the page 3 gate: everything outside the gate stays outside.

No clinical diagnosis

We do not diagnose depression, anxiety, or any disorder. **Diagnosis** means a clinical judgment about a disorder — that judgment belongs to clinicians, not to this study.

No individual labeling

We do not label individuals as “stressed” or “dissatisfied.” **Labeling** means attaching a fixed category to a person; we track patterns, not people.

No organization-level verdicts

We do not declare the organization healthy or unhealthy. A **verdict** is a final judgment — this study reports evidence, not verdicts.

No sharing of individual data

Individual responses never leave the research team. Only aggregated results are reported.

These four exclusions match the page 3 boundary and are re-checked in the page 28 audit.

Data handling & privacy

Where individual data stops being visible — and what leaves the research team.

INDIVIDUAL DATA STOPS HERE

01

Collection

Interviews & observation data, with names attached.

02

Anonymization

Names and identifiers are removed before analysis.

03

Analysis

Patterns are tested on anonymized data only.

04

Aggregated reporting

Results are reported in groups, not individuals.

Access Only the research team sees individual-level data; managers see **aggregated** (group-level) results.

Storage Data is stored securely and deleted after the study.

The boundary check protocol

A **protocol** — a repeated set of checks — not a one-time decision.

At collection

- Only in-boundary variables are collected: work demands, satisfaction expressions, stress cues.

At analysis

- No diagnostic labels are applied.
- Results stay **aggregated** (group-level, never individual).

At reporting

- No organization-level verdicts are drawn.
- Individual data is not shared.

Boundary check A **boundary check** is a verification that only the variables defined on page 3 are being used. The next page shows the result of that check.

Yes — the boundaries held

The gate on page 3 held.

Variables stayed inside
Only work demands, satisfaction expressions, and stress cues were collected.

No clinical diagnosis made
No diagnostic labels were applied at any stage.

No organization-level conclusions
All findings are framed as trackable evidence, not verdicts.

Individual data not shared
All reporting is **aggregated** — combined into group totals, so no single person can be identified.

Boundary — the line between what this study may touch and what it must not.

Respected — kept to, at every step, without exception.

One snapshot is not enough

One measurement



Snapshot — one moment in time. It shows where things stand now, not how they move.

vs

Repeated measurement



Pattern — the same evidence trail, extended in time. Satisfaction and stress change; the pattern is what matters, not the moment.

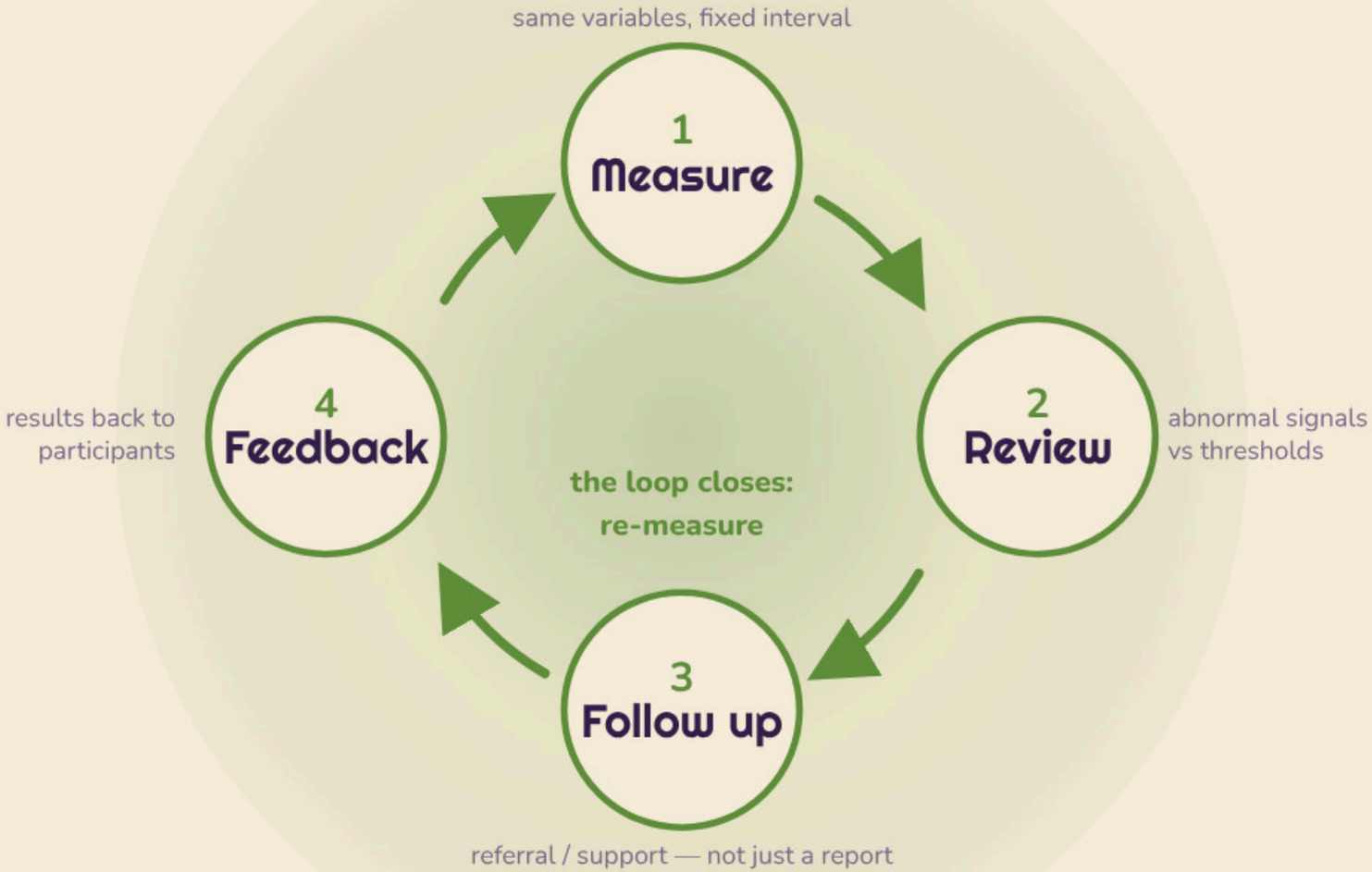
Repeated measurement

measuring the same variables again and again, at a fixed interval

turns snapshots into a trail — the same evidence trail we have been building, extended in time.

The monitoring loop

Loop — a process that repeats itself. **Cycle** — one full round of the loop, from start back to start.



Abnormal signal — a result that crosses a predefined threshold and needs attention.

Follow-up pathway — the concrete next step after a flagged signal: referral or support, not just a report.

Repeated measurement

The same variables, measured again at set intervals — the pattern matters more than any single point.



Illustrative — not data

Interval

The fixed gap between measurement rounds — the same questions, the same way, each time.

Wave

A variable rising and falling over time. We read the shape, not one point.

Stability

Some variables barely move; some shift. Both are information.

Test–retest reliability

Does the same person score similarly when asked again? Convention: 2 weeks to 2 months between administrations.

Pulse-survey practice (weekly / monthly / quarterly) is practitioner-level guidance, not peer-reviewed.

Reviewing abnormal signals

A signal is a value outside the expected range — set from the data, not from clinical cutoffs.



Illustrative — not data

Signal

A value that moves outside the expected range for that variable.

Threshold

The edge of the expected range — set from the data, not from clinical cutoffs.

Abnormal

Here: outside the expected range. Not a medical label.

Escalation

Passing a signal to the next step — here, the research team reviews it.

Follow-up pathway

A concrete next step after a flag — referral/support, not just a report.

Non-clinical: a signal triggers review and re-measurement — not a diagnosis. Signals are reviewed by the research team, not by managers or clinicians.

Evidence: Strudwick et al. (2023) — screening followed by referral to treatment improves mental health; feedback and advice alone do not.

Feedback to participants

After each measurement round — not only at the end — participants get their own pattern in plain language, plus next steps.



Feedback

What goes back to participants: their own pattern over time, in plain language, plus guidance on next steps.

Personal summary

A short report of your own satisfaction and stress over time.

Group summary

How the team's patterns look — aggregated, not individual labels.

The organization determines how feedback is provided, who provides it, and how follow-up with participants happens (OPM guidance).



THE LOOP CONTINUES

Satisfaction and dissatisfaction become trackable evidence — inside the boundaries we set.

What the roundtable does next: **agree on the variables**, **agree on the intervals**, **agree on the feedback formats.**